

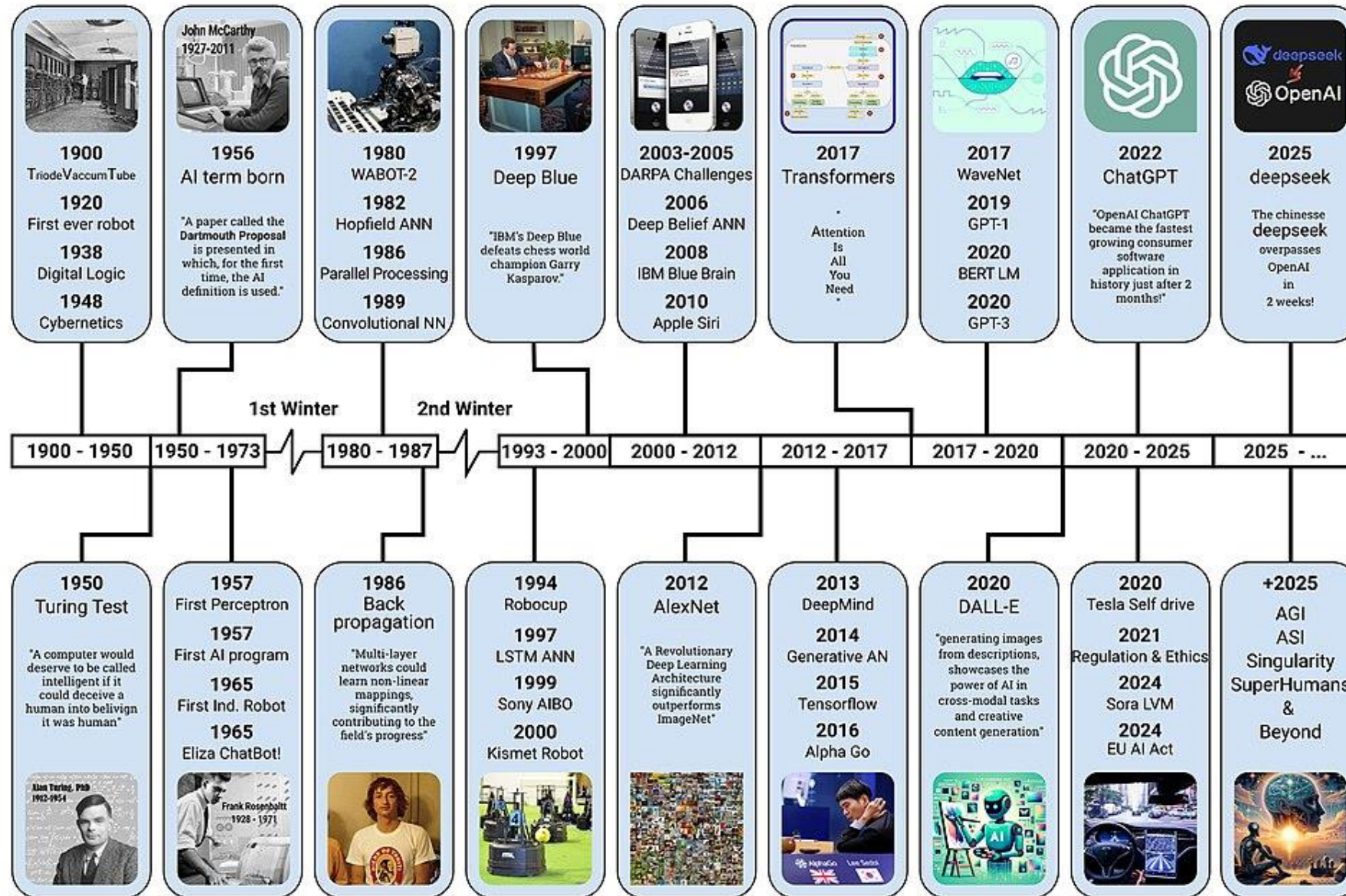
AIAA 2205: Introduction to AI

Enyan Dai

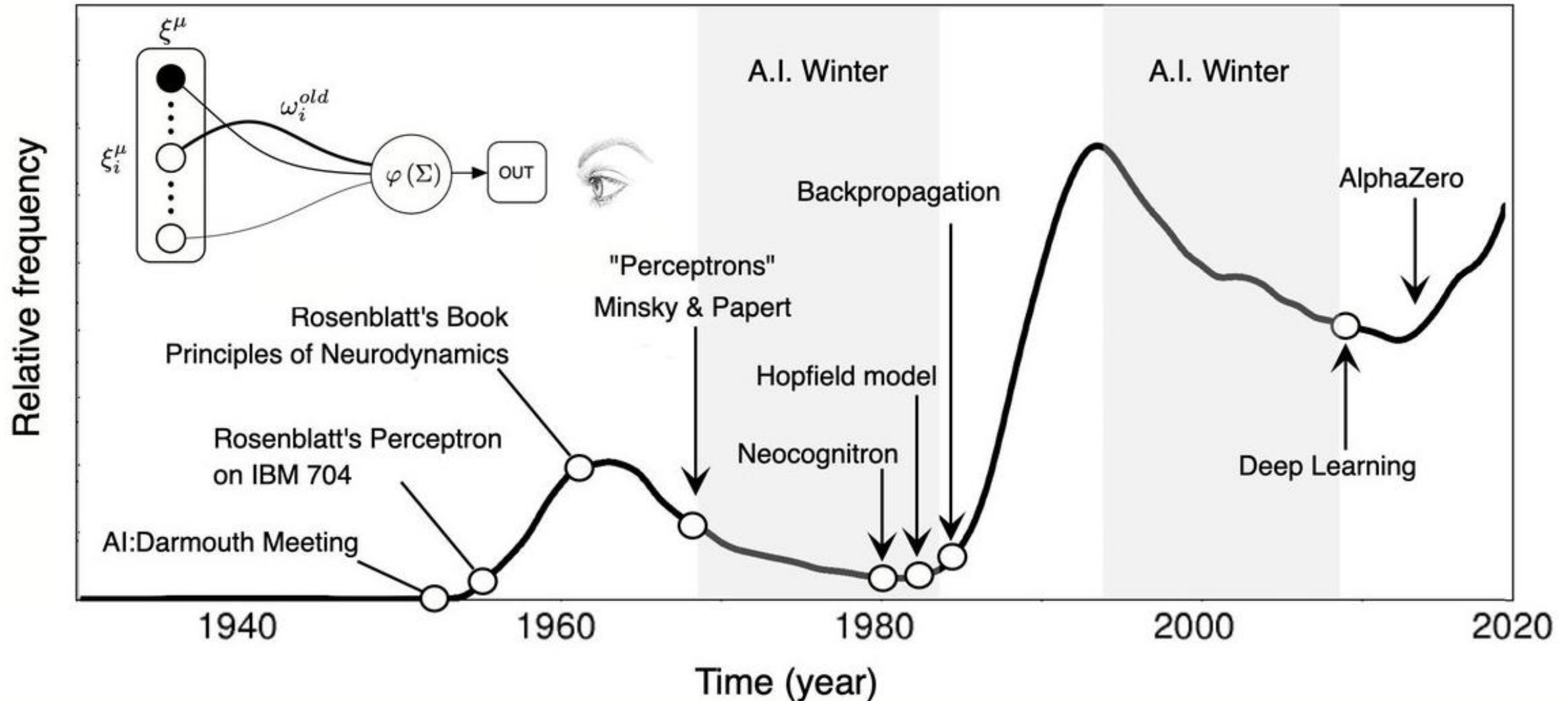
The Hong Kong University of Science and Technology (Guangzhou)

Recall of last lecture

Summary of AI History

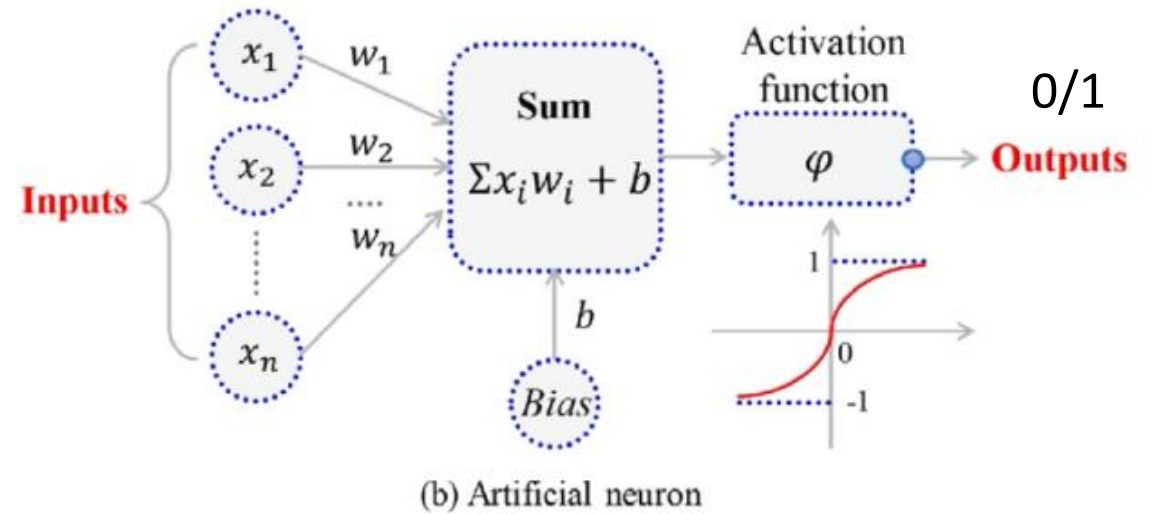
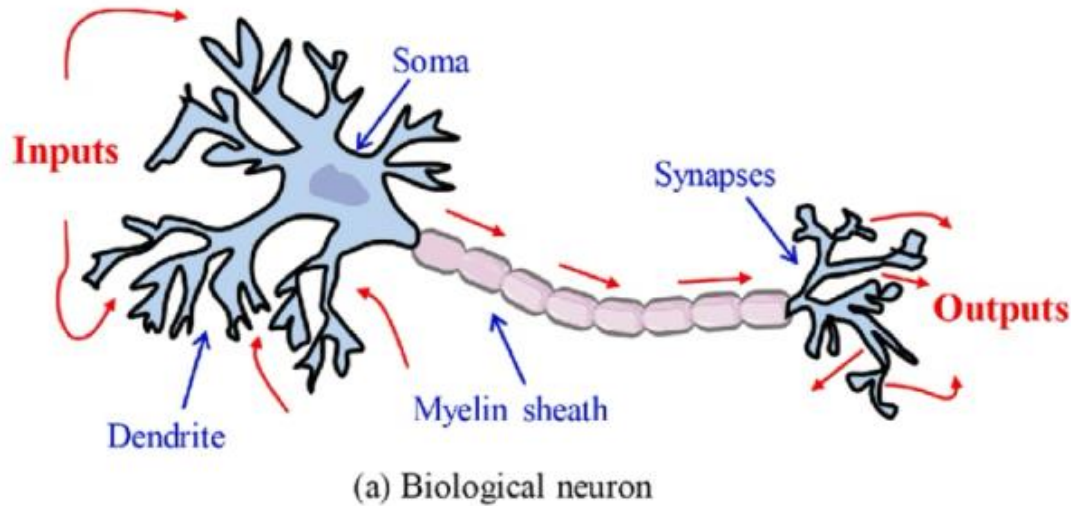


Development of artificial Neural Networks



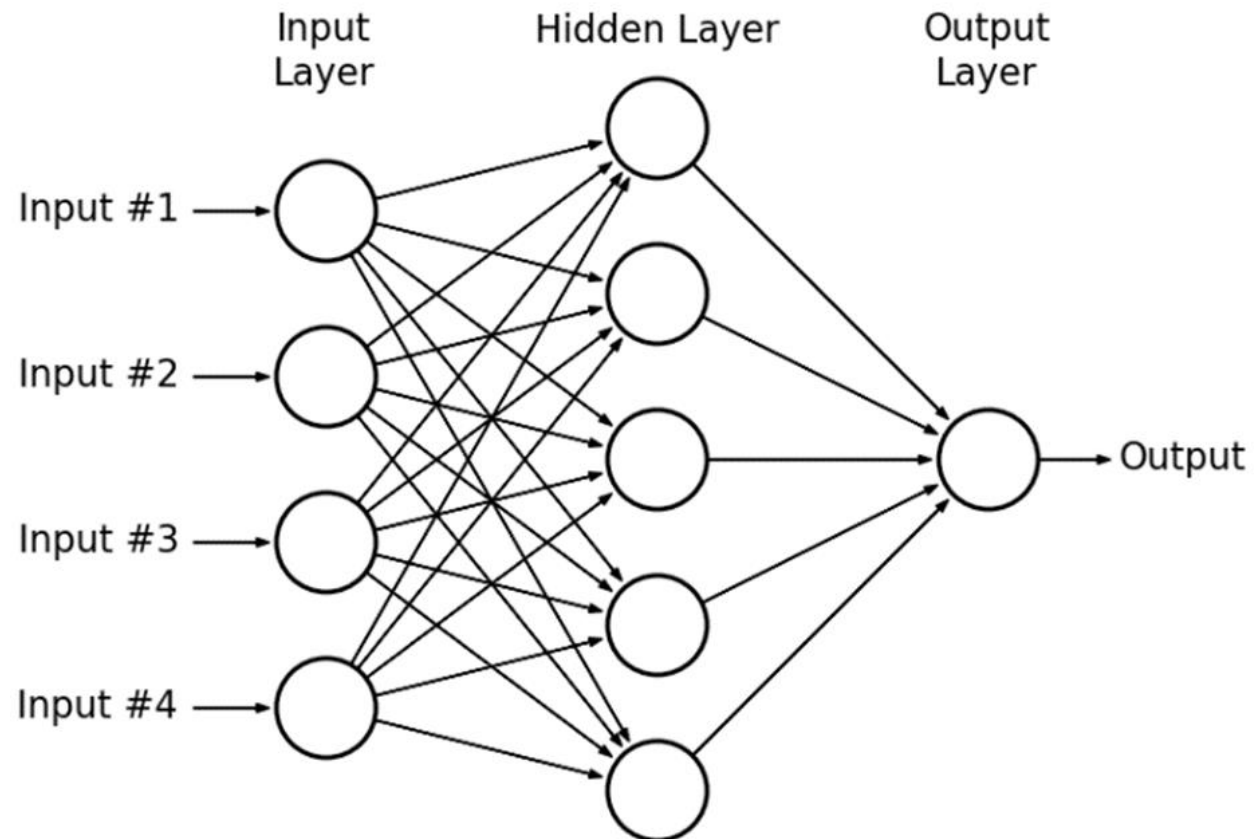
Perceptron

Inspired by the human neural network, the perceptron is developed from 1943 to 1958. Perceptron triggered widespread enthusiasm about the potential of ANN.



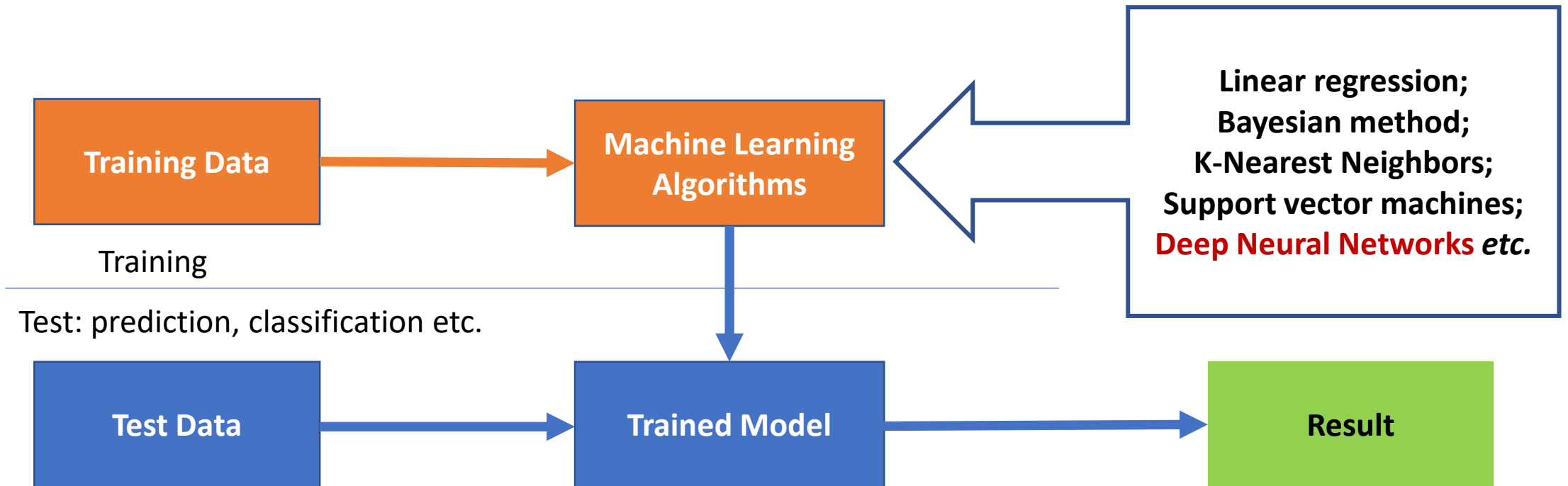
Multi-Layer Perception

- MLP Adds a hidden layer to solve the XOR problem

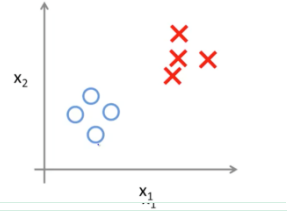


Current AI is Data-Driven AI

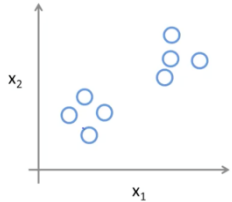
- Type of Artificial Intelligence that provides computers with the ability to **learn without being explicitly programmed**.
- **Various methods** can be used to make predictions based on data.



Machine Learning Approaches ?

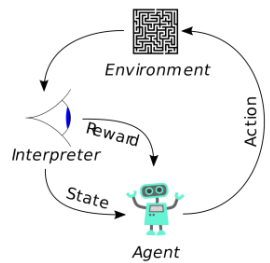


- **Supervised Learning:** Learning with a **labelled** training set
 - Example: email spam detector with training set of labelled emails



- **Unsupervised Learning:** Discovering patterns in **unlabelled** dataset
 - Example: cluster similar diseases based on the Medical Images

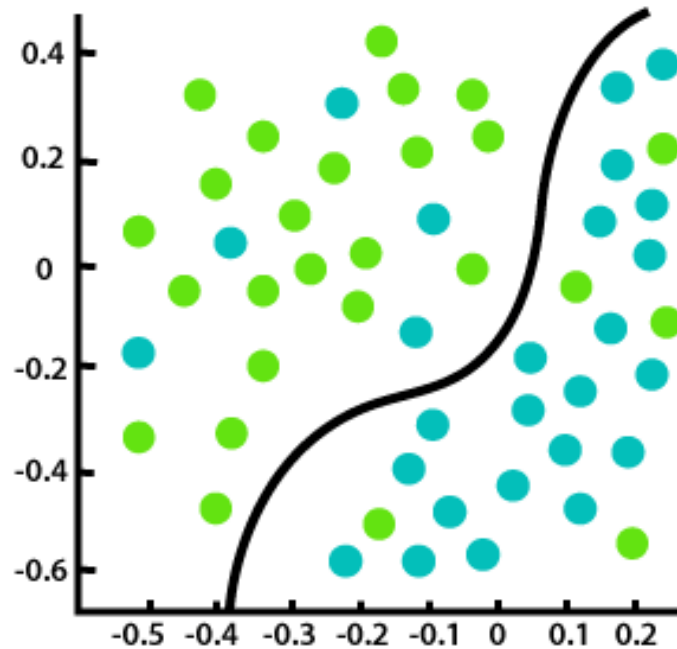
- **Semi-supervised Learning: Between** Supervised and Unsupervised Learning



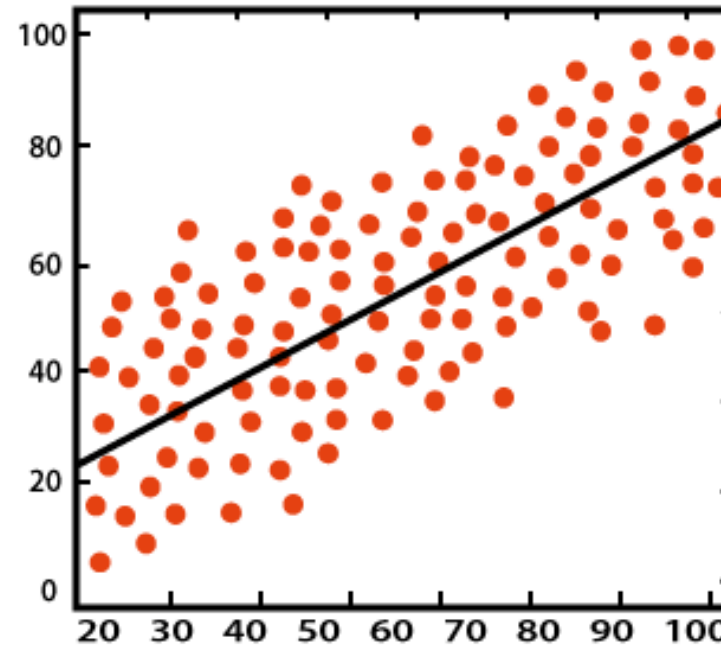
- **Reinforcement Learning:** Learning based on **feedback** or reward
 - Example: learn to play chess by winning or losing

Classification vs Regression

- **Classification:** used to predict/classify the **discrete** values such as Male or Female, True or False, Spam or not Spam, *etc.*
- **Regression:** used to predict the **continuous** values such as price, salary, *etc.*

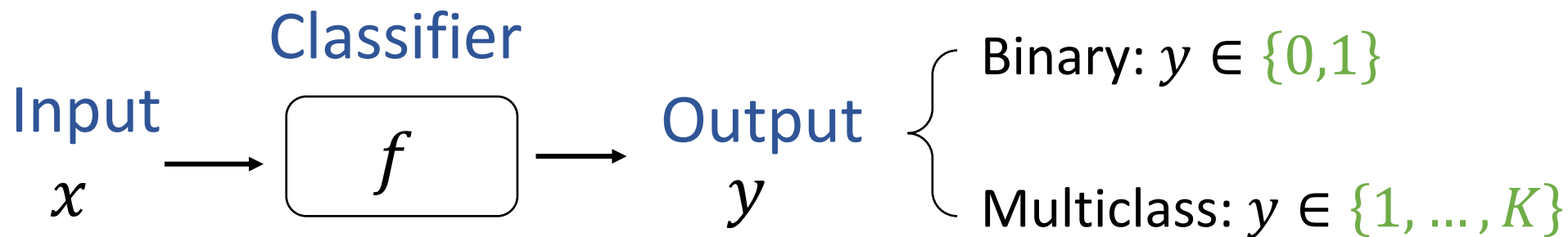


Classification



Regression

Classification



Spam Detection:



Email



<Spam or Not Spam>

Image Classification:

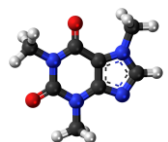


Image



<Cat, Dog,..., Bird>

Molecule Property Prediction:

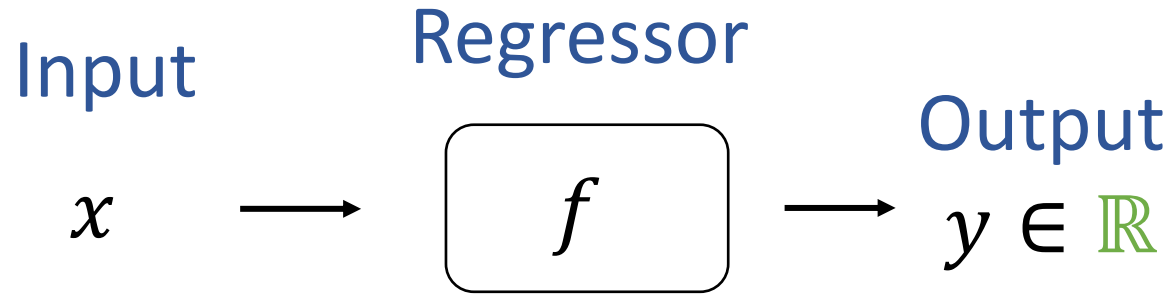


Molecule



<Mutagenic or Not Mutagenic>

Regression



House Price Prediction:  House Information → Price

Estimated Time of Arrival:  Traffic, Weather,... → Time of arrival

Question

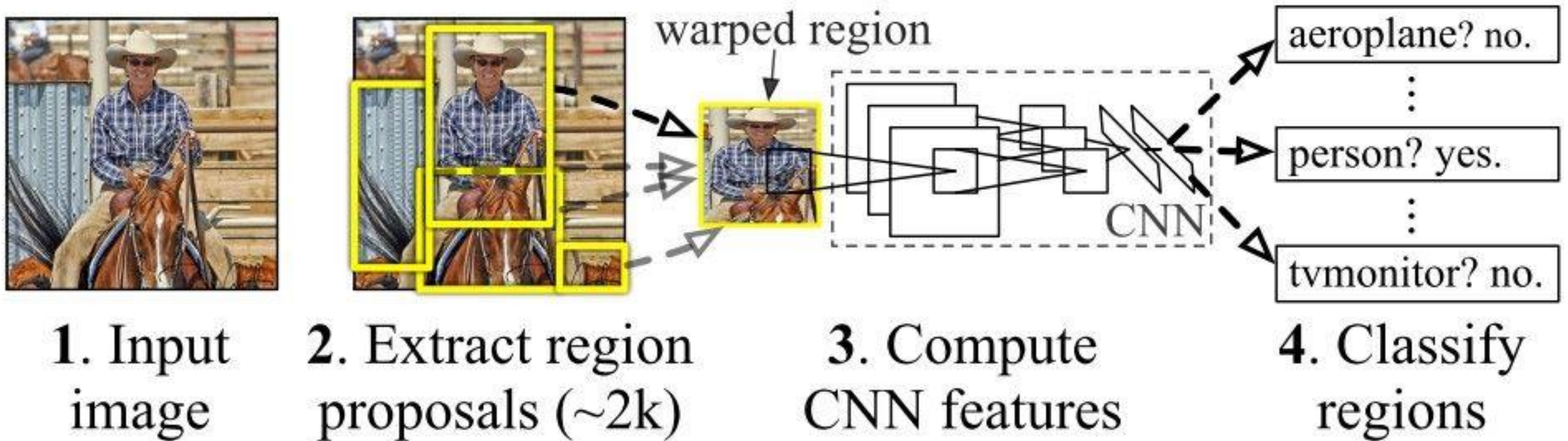
- **Object detection** aims get the bounding box of the objects such as person from the image
- Is object detection achieved by regression or classification?



Scan code to answer

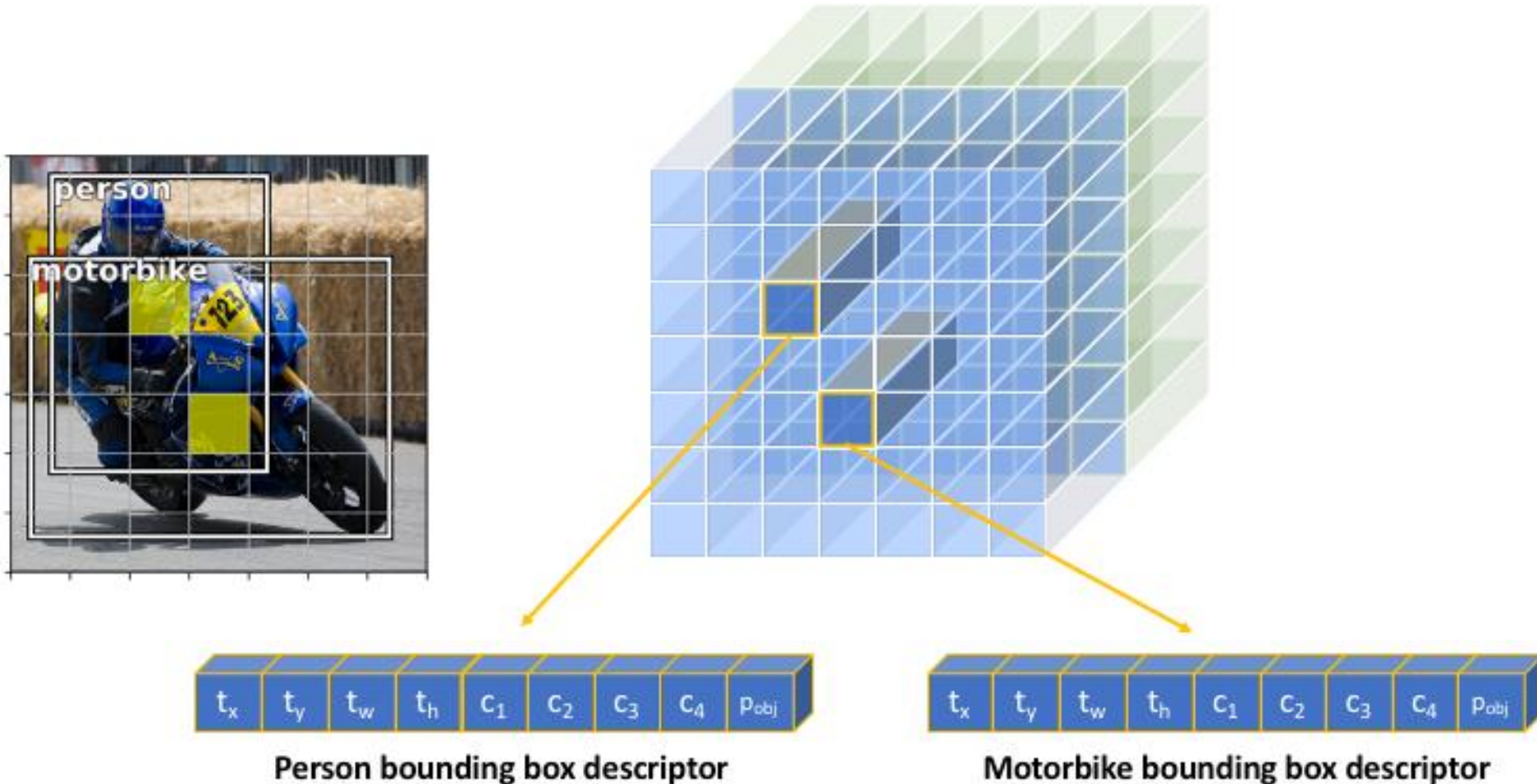
Object Detection

- Classical two-stage object detection is based on classification



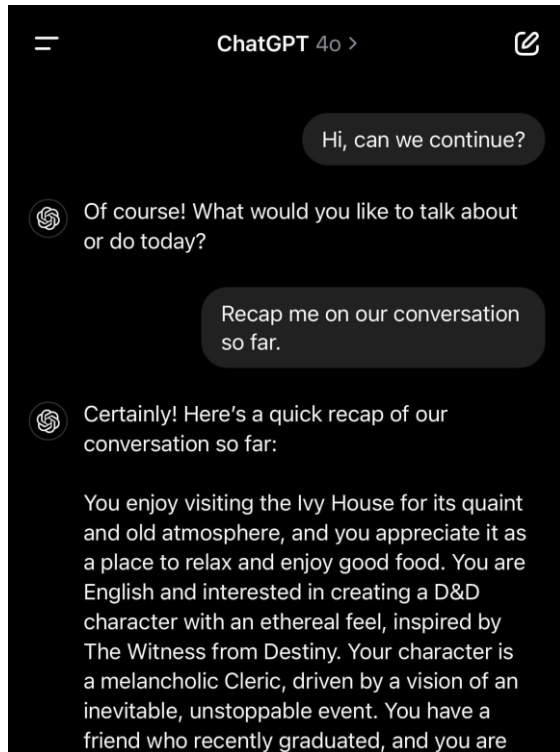
Object Detection

- One-stage object detection uses regressor to get bounding box



Large Language Model

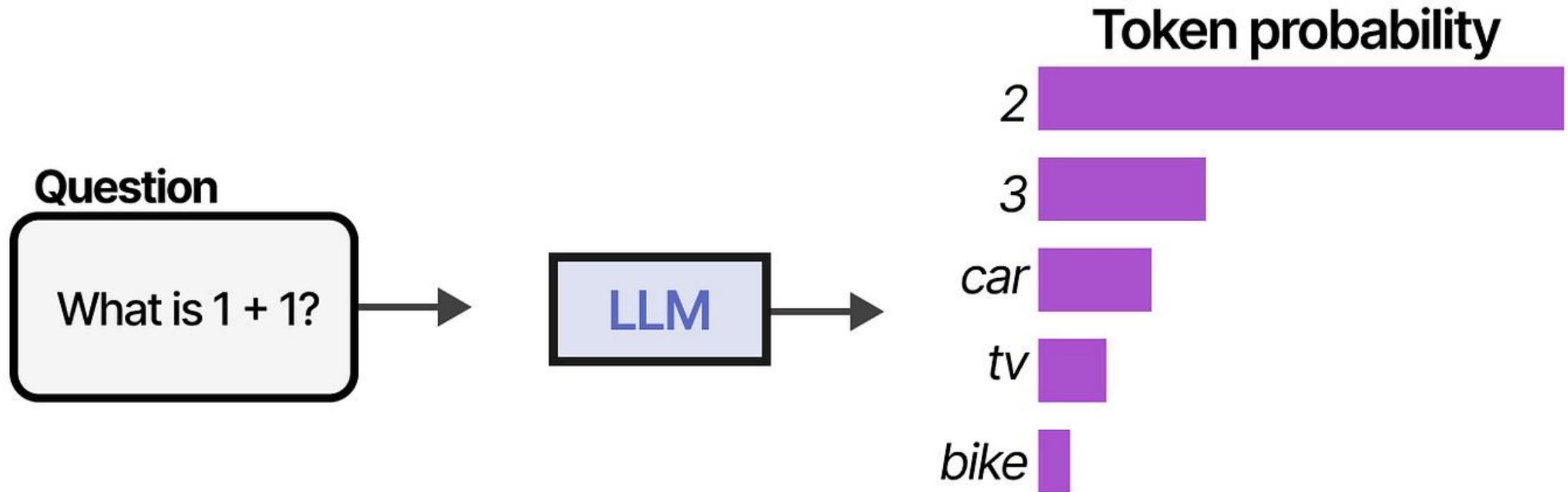
- Are LLMs such as ChatGPT and deepseek achieved by regression or classification?



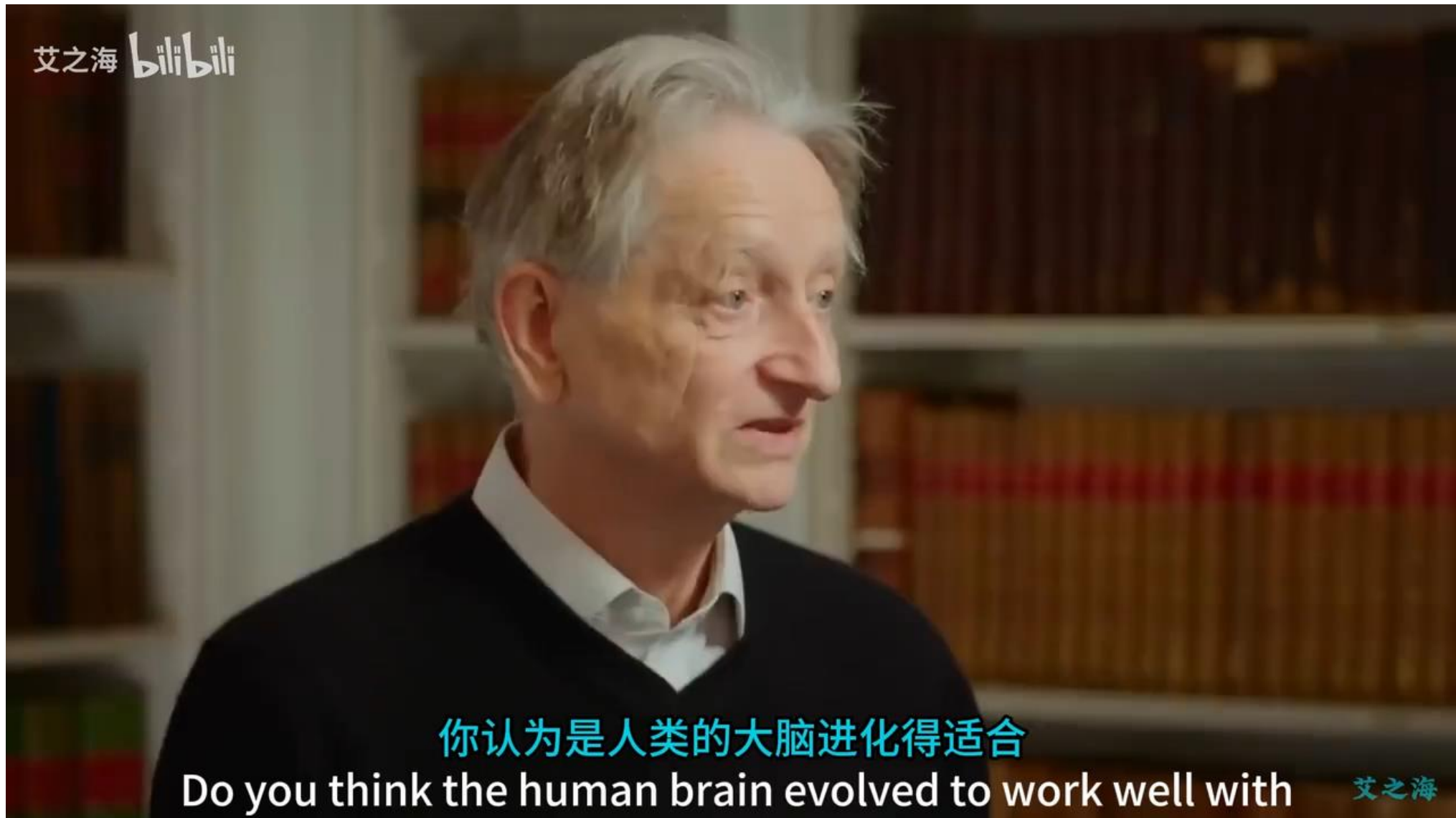
Scan code to answer

Large Language Model

- LLM is next token prediction, which is classification



Hinton Talks About Language





辛顿自曝成长史：
学不会数学，当过木匠
成就源自好奇心驱使

Linear Regression

The Discovery of Ceres



1801: astronomer Piazzi discovered Ceres, made 19 observations of location before it was obscured by the sun

Time	Right ascension	Declination
Jan 01, 20:43:17.8	50.91	15.24
Jan 02, 20:39:04.6	50.84	15.30
...	...	...
Feb 11, 18:11:58.2	53.51	18.43

When and where will Ceres be observed again?

Linear Regression



September 1801: Gauss took Piazzi's data and created a model of Ceres's orbit, makes prediction

December 7, 1801: Ceres located within $1/2$ degree of Gauss's prediction, much more accurate than other astronomers

Method: Least Squares Linear Regression

Regression

Buy **Rent** Transaction Estate New Property Carpark Agent Branch

Filtering

District | MTR | School Net

Estate

Phase

Rental

Saleable Area

Rooms

More

Reset

Sorting : Default

13,329 Properties for Rent



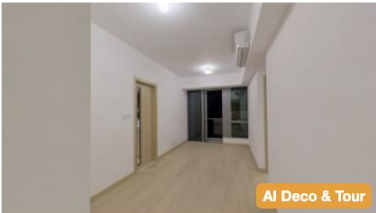
AI Deco & Tour

Park Central. Phase 3. Central Heights (Tower 13) Middle Floor. Flat D. 2 Rooms
Tseung Kwan O
16 years · North West · 75% Efficiency(%)
Exclusive Key Best Deal
S.A. **465 ft²** @ \$39 /ft² GFA **623ft²** @ \$29 /ft² Rent **\$ 18,000**



VR

Caribbean Coast 3 Rooms. 1 Suite
Tung Chung Town Centre
75% Efficiency(%)
Exclusive Key Best Deal
S.A. **927 ft²** @ \$25 /ft² GFA **1,228ft²** @ \$19 /ft² Rent **\$ 23,000**



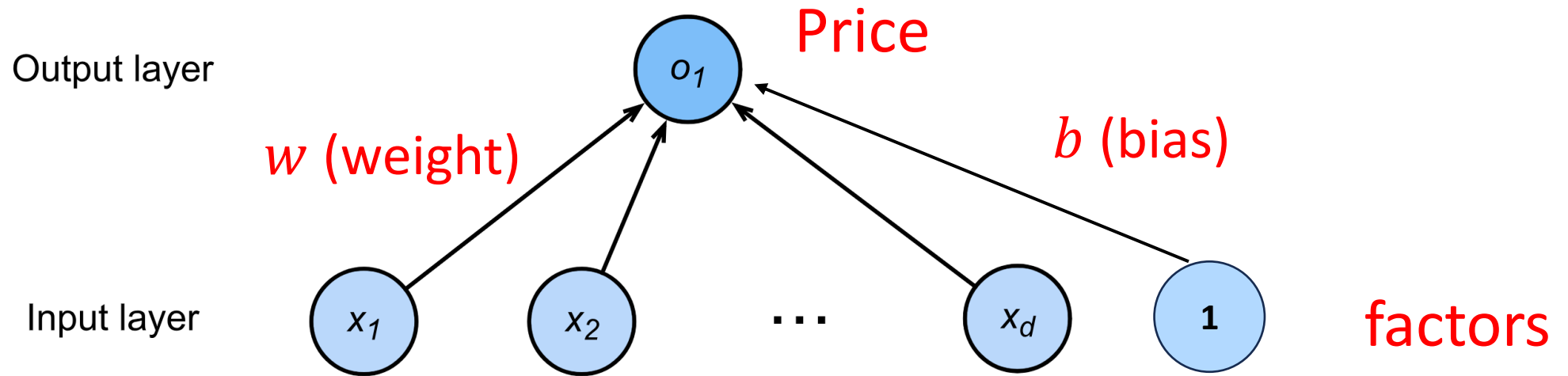
AI Deco & Tour

Lohas Park. Phase 7A Montara. Tower 2 (2A) High Floor. Flat E. 2 Rooms
Lohas Park
West
Exclusive Key
S.A. **559 ft²** Rent ~~20000~~ **\$ 18,500**

The price could be affected by

- Living area
- Number of rooms
- Construction time
- ...

Viewed as a single-layer neural network



$$f(\mathbf{x}) = \mathbf{w} \cdot \mathbf{x} = [w_1, \dots, w_d, b] \begin{bmatrix} x_1 \\ \vdots \\ x_d \\ 1 \end{bmatrix} = w_1 x_1 + \dots + w_d x_d + b$$

$\mathbf{x} \in \mathbb{R}^d$: Features of Example

$\mathbf{w} \in \mathbb{R}^d$: Parameters

Linear Regression

Given **training data**: $\mathcal{D} = \{(x_1, y_1), \dots, (x_N, y_N)\}$,
we aim to obtain a linear function:

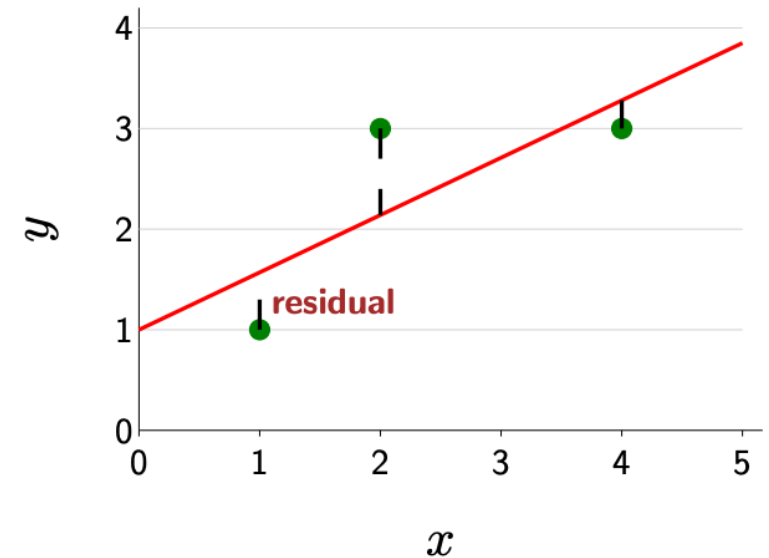
$$f(x) = \mathbf{w} \cdot \mathbf{x} + b,$$

where Mean Squared Error is minimized

	x	y
example	1	1
example	2	3
example	4	3

Mean Squared Error Loss:

$$\text{MSELoss}(\mathbf{w}) = \frac{1}{N} \sum_{i=1}^N (f(x_i) - y_i)^2$$



Solving w and b

Buy **Rent** Transaction Estate New Property Carpark Agent Branch

Filtering District | MTR | School Net Estate Phase Rental Saleable Area Rooms More Reset

Sorting: Default 13,280 Properties for Rent



Park Central. Phase 3. Central Heights (Tower 13) Middle Floor. Flat D. 2 Rooms
Tseung Kwan O
16 years - North West - 75% Efficiency(%)
Exclusive Key Best Deal
S.A. 465 ft² @ \$39 /ft² GFA 623ft² @ \$29 /ft² Rent \$ 18,000



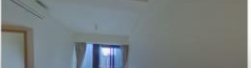
Caribbean Coast 3 Rooms. 1 Suite
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Lohas Park. Phase 7A Montara. Tower 2 (2A) High Floor. Flat E. 2 Rooms
Lohas Park
West
Exclusive Key
S.A. 559 ft² @ \$33 /ft² GFA 700ft² @ \$27 /ft² Rent \$ 18,500



Kingswood Villas. Phase 7 Kenswood Court 2 Rooms
Tin Shui Wai
North - 77% Efficiency(%)
Exclusive Key
S.A. 442 ft² @ \$24 /ft² GFA 576ft² @ \$18 /ft² Rent \$ 10,500



Mantin Heights. Tower 1 2 Rooms
Ho Man Tin
4 years - South
Key Best Deal
S.A. 442 ft² @ \$24 /ft² GFA 576ft² @ \$18 /ft² Rent \$ 10,500

What do we have?

Thousands of data for training.

$$D = \{(x_i, y_i) | i = 1, 2, \dots, n\}$$

Define the MSE loss function.

$$L = \sum_{i=1}^n (f(x_i; w, b) - y)^2$$

Loss function: how bad the model fits the data.

Picking w and b with minimal L .

$$\min_{w, b} \sum_{i=1}^n (f(x_i; w, b) - y)^2$$

Solving $\mathbf{w}$ and b

Picking $\mathbf{w}$ and b with minimal L .

$$\min_{\mathbf{w}, b} \sum_{i=1}^n (f(\mathbf{x}_i; \mathbf{w}, b) - y)^2$$

Rewrite in vector form: $\mathbf{x} = [x_1, x_2, \dots, x_n] \in R^{d \times n}$

$$\min_{\mathbf{w}, b} \left\| \mathbf{w}^T \mathbf{x} + b - \mathbf{y} \right\|^2$$

$$\mathbf{y} = [y_1, y_2, \dots, y_n] \in R^{1 \times n}$$

$$\mathbf{w} \in R^{d \times 1}$$

$$b \in R^1$$



Let $\mathbf{w}_a = [\mathbf{w}; b] \in R^{(d+1) \times 1}$, $\mathbf{x}_a = [\mathbf{x}; 1] \in R^{(d+1) \times n}$

A simplified formulation

$$\min_{\mathbf{w}_a} \left\| \mathbf{w}_a^T \mathbf{x}_a - \mathbf{y} \right\|^2$$

Solving $\mathbf{w}$ and b

A simplified formulation

$$\min_{\mathbf{w}_a} \left| \mathbf{w}_a^T \mathbf{x}_a - \mathbf{y} \right|^2$$

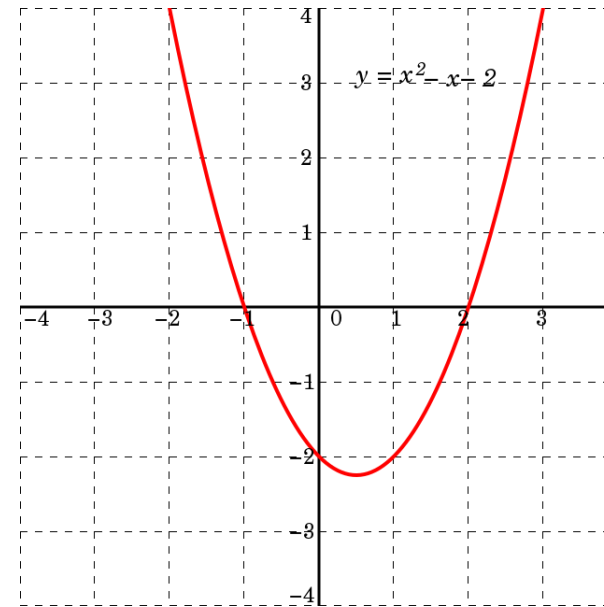
$$\begin{aligned} & \left| \mathbf{w}_a^T \mathbf{x}_a - \mathbf{y} \right|^2 \\ &= (\mathbf{w}_a^T \mathbf{x}_a - \mathbf{y})(\mathbf{w}_a^T \mathbf{x}_a - \mathbf{y})^T \\ &= \mathbf{w}_a^T \mathbf{x}_a \mathbf{x}_a^T \mathbf{w}_a + \mathbf{y} \mathbf{y}^T - \mathbf{y} \mathbf{w}_a^T \mathbf{x}_a - \mathbf{x}_a^T \mathbf{w}_a \mathbf{y} \\ &= \mathbf{w}_a^T \mathbf{x}_a \mathbf{x}_a^T \mathbf{w}_a + \mathbf{y} \mathbf{y}^T - 2 \mathbf{y} \mathbf{w}_a^T \mathbf{x}_a \end{aligned}$$

Solving w and b

A simplified formulation

$$\min_{w_a} \left\| w_a^T x_a - y \right\|^2$$

$$\begin{aligned} & \left\| w_a^T x_a - y \right\|^2 \\ &= (w_a^T x_a - y)(w_a^T x_a - y)^T \\ &= w_a^T x_a x_a^T w_a + y y^T - y w_a^T x_a - x_a^T w_a y \\ &= w_a^T x_a x_a^T w_a + y y^T - 2 y w_a^T x_a \end{aligned}$$



Quadratic function is a convex function.

Optimal solution $\longleftrightarrow$ Gradient is 0

Solving w and b

A simplified formulation

$$\min_{w_b} \left\| w_a^T x_a - y \right\|^2$$

$$\begin{aligned} & \left\| w_a^T x_a - y \right\|^2 \\ &= (w_a^T x_a - y)(w_a^T x_a - y)^T \\ &= w_a^T x_a x_a^T w_a + y y^T - y w_a^T x_a - x_a^T w_a y \\ &= w_a^T x_a x_a^T w_a + y y^T - 2 y w_a^T x_a \end{aligned}$$

Let $\frac{\partial L}{\partial w_a} = 0$, then we have

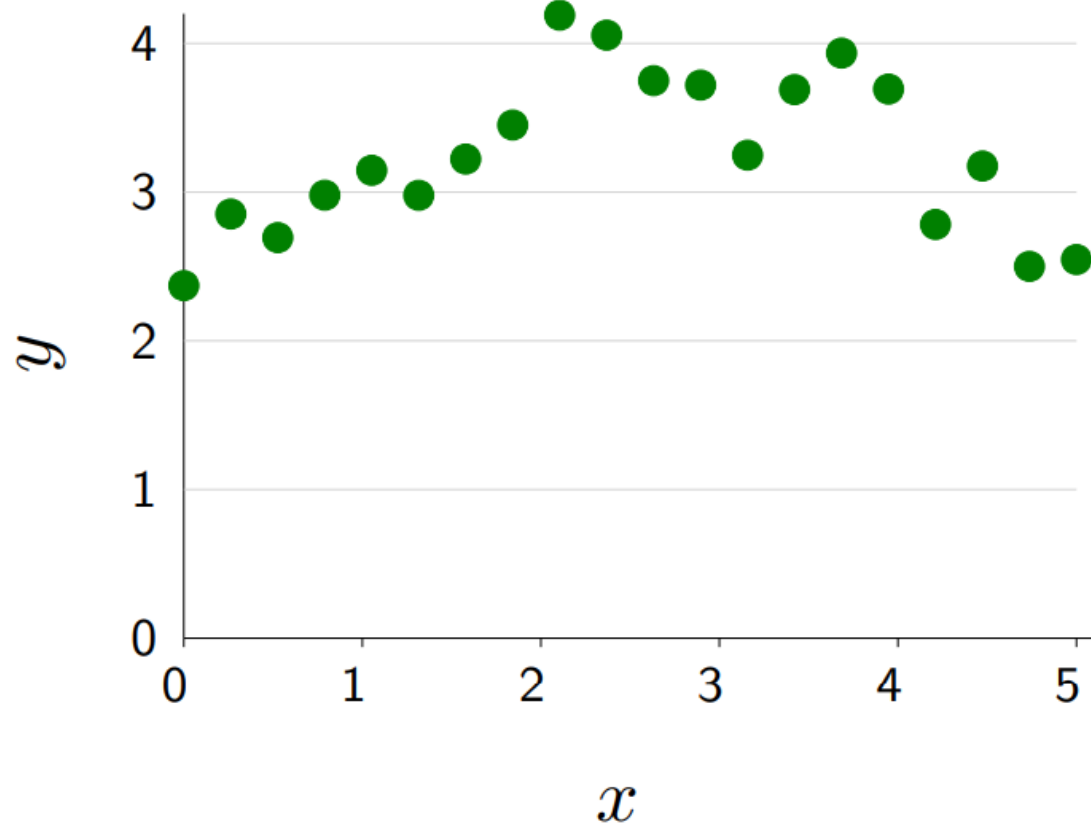
$$2 x_a x_a^T w_a - 2 x_a y = 0$$

A closed form solution

$$w_a = (x_a x_a^T)^{-1} x_a y$$

Complex Data

- How do we fit non-linear data?



None-linear Features

- Polynomial function

$$\phi(x) = [1, x, x^2]$$

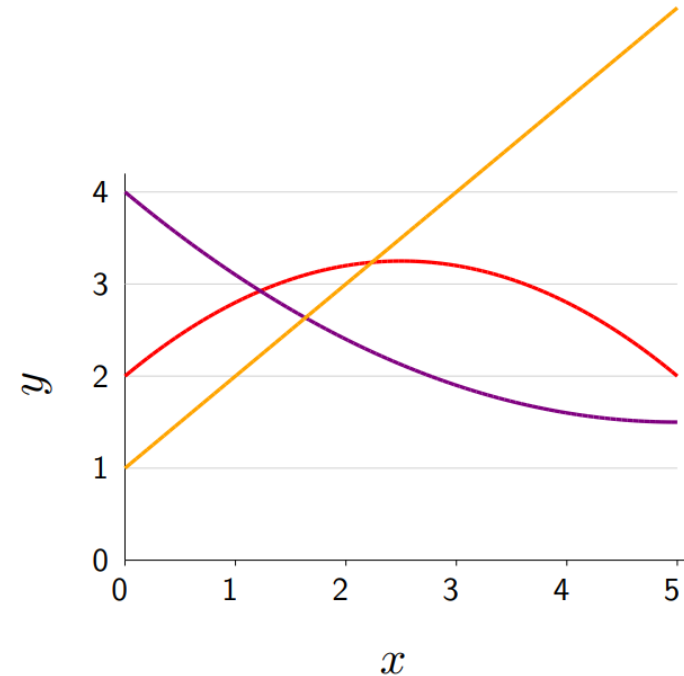
Example: $\phi(3) = [1, 3, 9]$

$$f(x) = [2, 1, -0.2] \cdot \phi(x)$$

$$f(x) = [4, -1, 0.1] \cdot \phi(x)$$

$$f(x) = [1, 1, 0] \cdot \phi(x)$$

$$\mathcal{F} = \{f_{\mathbf{w}}(x) = \mathbf{w} \cdot \phi(x) : \mathbf{w} \in \mathbb{R}^3\}$$



None-linear Features

- Cos function and any other you can think about

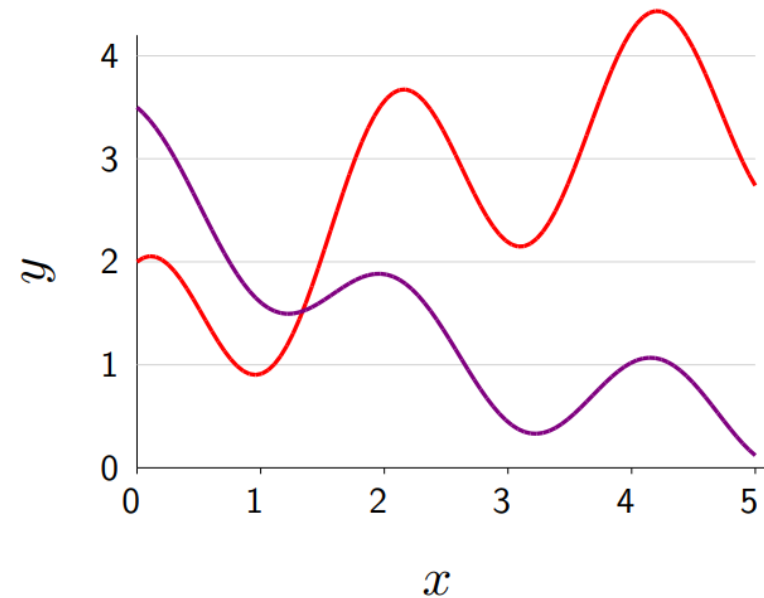
$$\phi(x) = [1, x, x^2, \cos(3x)]$$

$$\text{Example: } \phi(2) = [1, 2, 4, 0.96]$$

$$f(x) = [1, 1, -0.1, 1] \cdot \phi(x)$$

$$f(x) = [3, -1, 0.1, 0.5] \cdot \phi(x)$$

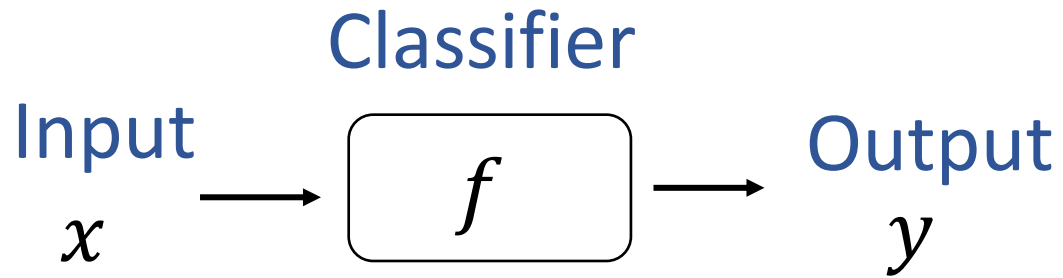
$$\mathcal{F} = \{f_{\mathbf{w}}(x) = \mathbf{w} \cdot \phi(x) : \mathbf{w} \in \mathbb{R}^4\}$$



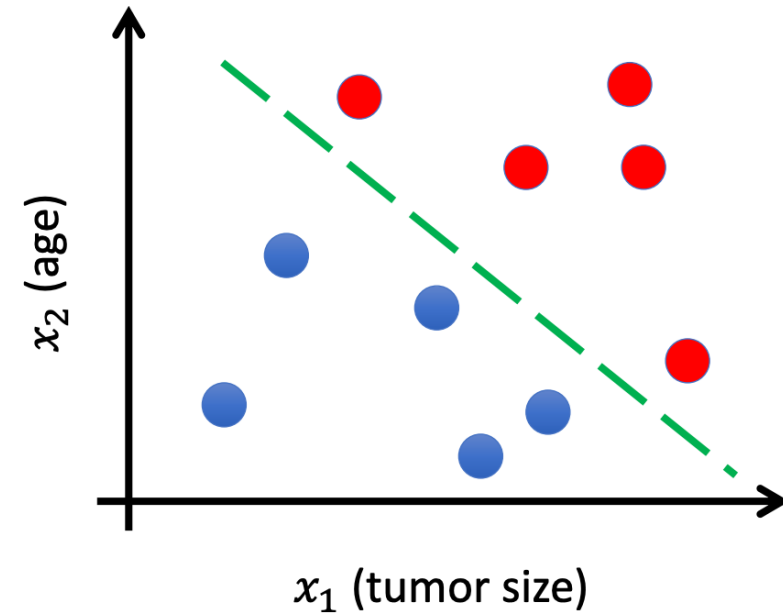
For a very long period, **feature engineering** plays a crucial role

Logistic Regression

Logistic regression is used for **Binary Classification!**



- Binary: $y \in \{0, 1\}$
- Multiclass: $y \in \{1, \dots, K\}$



Example: **Malignant** vs. **Benign** Ocular Tumor

Logistic Regression

Sample	Probability to Estimate	Ground Truth
x_1	$P(y = 1 x_1) = p_1$	$P(y = 1 x_1) = 1$
x_2	$P(y = 1 x_2) = p_2$	$P(y = 1 x_1) = 0$

Adopt linear regressor $w \cdot x$ to estimate p ?

NO!!! Because $p \in [0,1]$

Solution: Adopt $w \cdot x$ to regress the logits of the event:

$$w \cdot x = \log \left(\frac{p}{1-p} \right) \in (-\infty, +\infty)$$

Logistic Regression

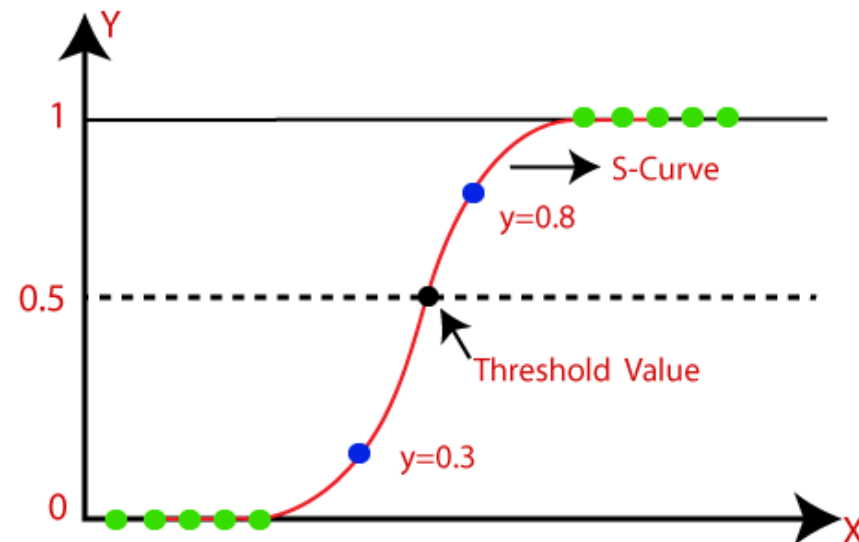
Estimate the Logit:

$$\mathbf{w} \cdot \mathbf{x} = \log\left(\frac{p}{1-p}\right) \Rightarrow f(\mathbf{x}) = \frac{1}{1 + \exp(-\mathbf{w} \cdot \mathbf{x})} = p$$

Sigmoid Function $\sigma(z) = \frac{1}{1 + \exp(-z)}$

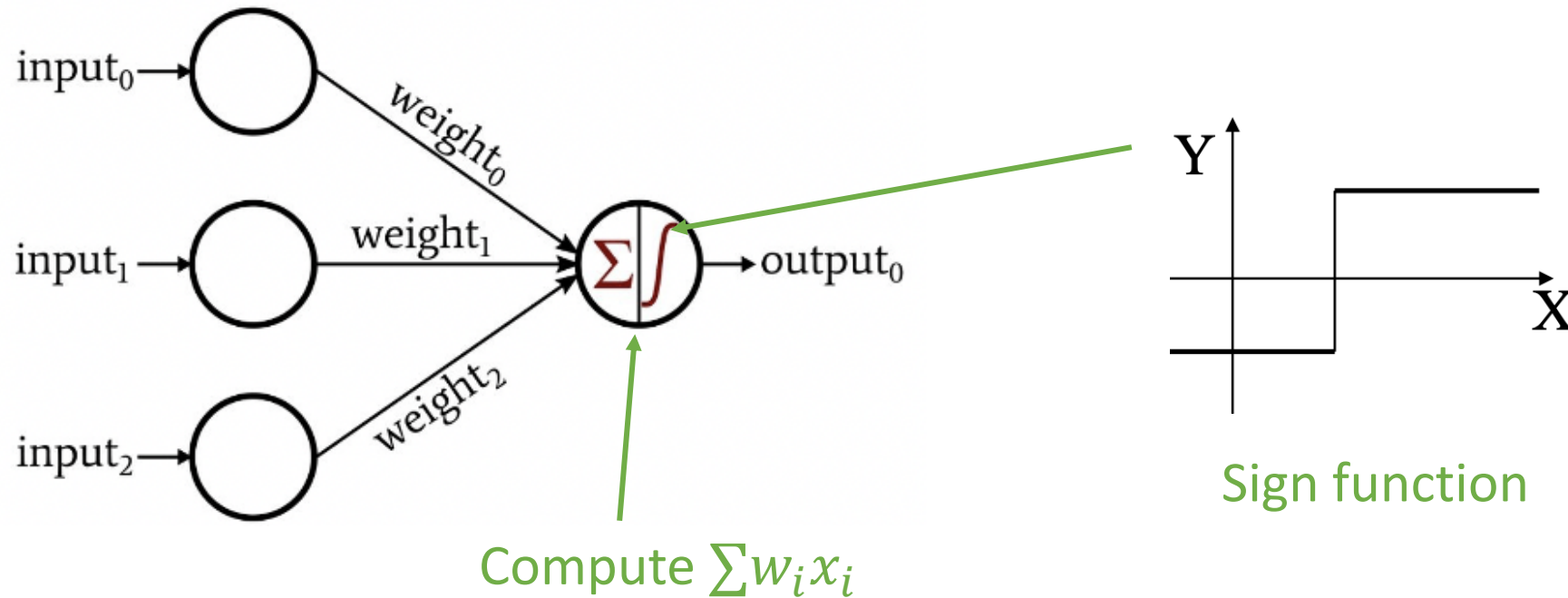
Decision Boundary: $\mathbf{w} \cdot \mathbf{x} = 0$

Optimization: Gradient Descent
with Cross Entropy Loss



Perception

- Perceptron is can be viewed as a logistics regression
- Equation: $y = \text{sign}(\sum w_i x_i)$



How to optimize it?